Chapter

Cell Cycle and Structure of Chromosomes

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1. Duplicated chromosomes are joined at a point termed:

(a) Centrosome

(b) Centromere

(c) Centriole

(d) Chromatid **MAIN 2024**

2. A female human cell divides through mitosis, maintaining the same number of chromosomes. Consider the chromosome composition of the daughter cells after mitosis. After Mitosis, a female human cell will have :

(a) 44+ XX chromosomes.

(b) 22+ X chromosomes.

(c) 22+ Y chromosomes.

(d) 44+ XY chromosomes. **COMP 2021**

3. At the end of ___________ Cytokinesis is completed.

(a) Metaphase

(b) Prophase

(c) Interphase

(d) Telophase **SQP 2021**

4. The nitrogenous base Adenine always pairs with __________

(a) Thymine

(b) Guanine

(c) Cytosine

(d) Thiamine **MAIN 2021**

5. Replication of DNA in the cell cycle occurs during the :

(a) G1 – phase

(b) Anaphase

(c) S - phase

(d) G_2- phase **MAIN 2021**

6. Which phase comes between G_1 and G_2 phase?

(a) G. phase

(b) M - phase

(c) S- phase

(d) Interphase **MAIN 2021**

7. The stage in which daughter chromosomes move toward the poles of the spindle is:

(a) Anaphase

(b) Metaphase

(c) Prophase

(d) TeloPhase

8. State the functions of Centromere:

(a) It is the point of attachment of two sister chromatids.

(b) It is the point of attachment of two centrioles.

(c) It is the point of attachment of two centrosomes.

(d) It is the point of attachment between two daughter nuclei. **COMP 2021**

9. Chrosomes get aligned at the centre of the cell during ____________

(a) Metaphase
(b) Anaphase

(c) Prophase
(d) Telophase **COMP 2022**

10. The type of cell division that leads to the formation of two identical daughter cells is

(a) Mitosis

(b) Genetic recombination

(c) Meiosis

(d) Cytokinesis **SQP 2021**

11. In complementary base pairing of DNA, guanine always pairs with ___________ and adenine always pairs with ____________

(a) Cytosine, uracil

(b) Cytosine, thymine

(c) Thymine, cytosine

(d) Guanine, adenine **COMP 2021**

12. Chromosomes

(a) The carriers of heredity

(b) The controlling centre of the cell

(c) The site for various chemical reactions

(d) Intracellular digestion. **SQP 2023**

13. Gene

(a) Physical and functional unit of heredity that carries DNA material from one generation to the next.

(b) Physical and functional unit of heredity that carries chromosomes from one generation to the next.

(c) Physical and functional unit of heredity that carries gametes from one generation to the next.

(d) Physical and functional unit of heredity that carries genetic information from one generation to the next. **COMP 2022**

14. Assertion: Every chromosome, during metaphase has two chromatids.

Reason: Synthesis of DNA takes places in the S-phase of interphase.

(a) If both assertion and reason are true and reason is the correct explanation of assertion.

(b) If both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) If assertion is true but reason is false.

(d) If both Assertion and Reason are false. **MAIN 2022**

15. A nitrogenous base in DNA. MAIN 2022

16. The two kinds of cell division found in living organisms. COMP 2023

17. A membrane that disappears during late prophase SQP 2023

18. The specific part of a chromosome that determines hereditary characteristics. COMP 2021

19. Four daughter cells are formed as a result of meiosis. COMP 2021

20. The resting stage in mitosis is called interphase. MAIN 2023

21. The chromosome number is______in meiosis. COMP 2021

22. How many meiotic divisions are necessary to produce 100 pollen grains ? COMP 2022

23. During Mitosis what is the position of chromatids in: SQP 2025

(a) Metaphase
(b) Anaphase

24. State how does meiosis maintain the chromosome number in a species?
COMP 2023

25. How does cytokinesis occur in an animal cell? SQP 2022

Solution

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1. (b) Centromere

Duplicated chromosomes consist of two identical sister chromatids joined together at a point called the **centromere**.

2. (a) 44 + XX

A female human cell has **46 chromosomes (44 autosomes + XX sex chromosomes)**.
During **mitosis**, the daughter cells formed are **genetically identical** to the parent cell, maintaining the same chromosome number and composition.
Hence, each daughter cell will have **44 + XX chromosomes**.

3. (d) Telophase

Cytokinesis finishes by the end of telophase, forming two daughter cells.

4. (a) Thymine

In DNA, **Adenine** pairs with **Thymine** by **two hydrogen bonds**, following the base-pairing rule (A–T and G–C).

5. (c) S phase

During the **S (Synthesis) phase** of the cell cycle, **DNA replication** takes place, resulting in the duplication of chromosomes.

6. (c) S phase

The **interphase** consists of **G_1, S, and G_2 phases**, where the **S (Synthesis) phase** lies between G_1 and G_2 and is the stage of **DNA replication**.

7. (a) Anaphase

In **anaphase**, the **centromeres split**, and the **sister chromatids** separate and move to **opposite poles** as the **spindle fibers shorten**.

8. (a) **It is the point of attachment of two sister chromatids.**

The **centromere** joins the **two sister chromatids** of a duplicated chromosome and provides the **attachment site for spindle fibers** during cell division to ensure proper chromosome segregation.

9. (a) Metaphase

During **metaphase**, chromosomes attach to the **spindle fibers** by their **centromeres** and align along the **equatorial plane** (centre) of the cell.

10. (a) Mitosis

Mitosis is the process by which a parent cell divides to form **two genetically identical daughter cells**, each having the same number of chromosomes as the parent cell.

11. (b) Cytosine, thymine

In DNA, **guanine pairs with cytosine** through **three hydrogen bonds**, and **adenine pairs with thymine** through **two hydrogen bonds** — a rule known as **complementary base pairing**.

12. (a) The carriers of heredity

Chromosomes carry **genes**, which are responsible for transmitting **hereditary traits** and **genetic information** from parents to offspring.

13. (d) Physical and functional unit of heredity that carries genetic information from one generation to the next.

A **gene** is the **basic unit of heredity**, consisting of a specific sequence of nucleotides on a chromosome that carries **genetic information** needed to produce proteins and pass traits to the next generation.

14. (a) If both assertion and reason are true and reason is the correct explanation of assertion.

During the **S-phase**, DNA replication occurs, forming **two identical chromatids** joined at the **centromere**.

Hence, in **metaphase**, each chromosome appears as **two chromatids**, making the reason the correct explanation of the assertion.

15. Adenine

16. Mitosis and Meiosis

17. Nuclear membrane.

18. DNA

19. True.

20. True

21. Halved

22. 25 meiotic divisions.

23. (a) In the **equatorial plane**
(b) **Sister chromatids separate and move towards opposite poles**

24. During **meiosis**, gametes are formed with **half the number of chromosomes** (haploid). When the **male and female gametes fuse** during fertilization, the **diploid chromosome number** is restored. Thus, meiosis maintains the chromosome number in a species.

25. In an animal cell, a **cleavage furrow** forms in the plasma membrane during **telophase**. The furrow **deepens inward** and finally **divides the cell into two daughter cells**.